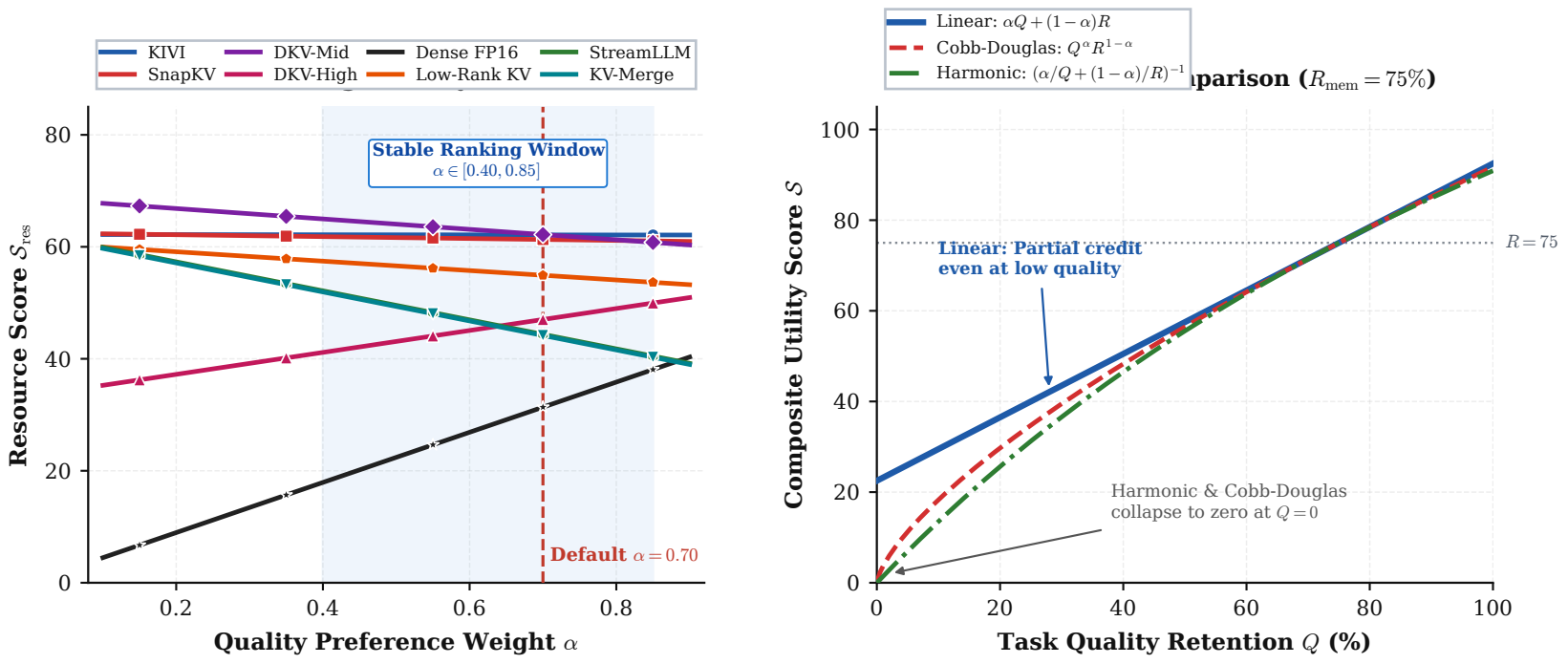
KIVI
SnapKV
DKV-Mid
DKV-High
Dense FP16
Low-Rank KV
StreamLLM
KV-Merge
Stable Ranking Window
$\alpha \in [0.40, 0.85]$
Default $\alpha = 0.70$
Resource Score $\mathcal{S}_{\text{res}}$
Quality Preference Weight α
Linear: $\alpha Q + (1-\alpha)R$
Cobb-Douglas: $Q^{\alpha}R^{1-\alpha}$
Harmonic: $(\alpha/Q + (1-\alpha)/R)^{-1}$
parison ($R_{\text{mem}} = 75\%$)
Linear: Partial credit
even at low quality
$R = 75$
Harmonic & Cobb-Douglas
collapse to zero at $Q = 0$
Composite Utility Score $\mathcal{S}$
Task Quality Retention Q (%)